

LUNGO, Sweden

WMO: 023820

Lat: 62.642N

Lon: 18.090E

Elev: 59

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.6	5.3	-4.3	4.4	0.9	0.3	5.6	5.9	43.0	22.0	38.7	25.8	11.2	270	0.762

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	73.7	64.7	70.6	63.1	68.1	61.6	67.3	71.6	64.9	69.0	62.8	66.5	11.2	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	93.9	70.3	63.1	86.6	67.5	61.1	80.6	65.1	31.8	71.8	30.0	69.3	28.4	66.6	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.1	31.3	27.2	DB	-2.8	79.3	8.6	3.6	-9.0	81.9	-14.0	84.0	-18.8	86.0	-25.1	88.6	(4)
(5)			WB	-3.2	68.0	7.1	2.7	-8.3	69.9	-12.5	71.5	-16.5	73.0	-21.6	75.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.9	26.1	25.2	30.6	37.6	46.4	55.2	61.7	60.2	52.4	42.5	35.1	29.1
	DBStd	14.25	8.82	9.97	7.53	4.93	5.39	4.77	4.51	4.43	4.83	5.39	6.65	8.67
	HDD50	3924	742	694	601	373	139	9	0	0	31	237	449	649
	HDD65	8454	1207	1114	1066	822	577	294	126	163	377	696	898	1114
	CDD50	980	0	0	0	0	28	166	362	315	104	5	0	0
	CDD65	36	0	0	0	0	0	1	23	12	0	0	0	0
	CDH74	81	0	0	0	0	3	2	58	18	0	0	0	0
	CDH80	7	0	0	0	0	1	0	5	1	0	0	0	0
Wind	WSAvg	12.3	13.9	13.4	12.4	11.3	11.2	11.2	10.8	11.2	11.7	12.9	13.7	13.9
Precipitation	PrecAvg	28.2	2.1	1.6	1.8	1.8	1.8	1.9	2.4	3.0	3.1	3.0	3.5	2.5
	PrecMax	42.2	5.2	4.6	5.0	5.6	5.3	5.7	6.5	7.8	8.3	8.9	10.2	10.9
	PrecMin	17.5	0.0	0.2	0.1	0.1	0.5	0.2	0.2	0.5	0.2	0.3	0.7	0.2
	PrecStd	5.2	1.3	1.0	1.1	1.3	1.0	1.1	1.5	1.9	1.7	1.9	2.2	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.4	45.1	51.6	56.4	67.1	73.3	79.0	76.3	67.3	56.7	51.9	45.5
		MCWB	39.0	39.3	42.9	46.2	54.3	58.8	66.5	67.1	57.0	51.1	46.6	41.4
	2%	DB	39.6	42.6	46.5	51.3	61.1	68.7	75.3	72.2	63.6	53.8	46.7	42.2
		MCWB	36.9	37.9	40.0	43.4	52.0	57.6	66.2	65.2	57.2	50.8	44.7	39.4
	5%	DB	38.0	39.3	43.0	48.2	57.8	65.5	72.0	69.5	61.2	52.1	44.8	40.4
		MCWB	35.9	35.5	37.5	41.6	49.6	57.0	64.4	63.5	56.0	49.6	43.3	38.4
	10%	DB	36.4	36.6	40.2	45.2	54.7	63.0	69.1	67.2	59.3	50.3	43.3	39.0
		MCWB	34.8	34.2	35.7	40.1	48.0	55.4	62.9	61.7	55.4	48.0	42.0	37.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.6	40.3	43.7	47.0	56.7	62.1	70.5	69.4	60.8	53.7	47.2	42.1
		MCDB	41.1	44.2	49.9	54.1	64.2	67.7	76.1	73.0	63.7	54.8	50.1	44.3
	2%	WB	37.8	38.2	40.4	44.5	53.1	60.0	68.1	67.0	59.1	51.9	45.1	40.2
		MCDB	39.0	41.9	45.7	50.0	59.5	65.5	72.7	70.7	61.3	53.0	46.4	41.6
	5%	WB	36.4	36.1	38.0	42.5	50.8	58.3	65.9	64.9	57.7	50.2	43.8	38.9
		MCDB	37.5	38.3	42.2	47.2	56.2	63.7	70.6	68.4	59.9	51.7	44.7	40.1
	10%	WB	34.9	34.6	36.2	40.6	48.7	56.7	63.7	62.9	56.3	48.2	42.2	37.5
		MCDB	36.2	36.3	39.4	44.4	53.3	61.7	68.0	66.1	58.5	50.0	43.1	38.7
Mean Daily Temperature Range	5% DB	MDBR	8.2	9.3	10.7	10.5	11.3	11.9	11.0	10.8	9.7	8.0	6.5	7.8
		MCDBR	7.6	10.9	13.5	16.2	16.4	15.5	15.3	14.2	12.1	9.5	7.1	8.0
		MCWBR	6.7	8.1	8.7	9.8	9.2	8.2	8.6	8.4	7.5	7.0	6.1	7.0
	5% WB	MCDBR	7.0	9.8	12.3	15.3	15.1	13.3	13.2	12.2	9.9	7.8	6.5	7.5
		MCWBR	6.5	7.7	8.3	9.7	9.3	7.9	8.0	7.9	7.1	6.7	6.0	6.9
Clear-Sky Solar Irradiance	taub	0.240	0.297	0.318	0.355	0.355	0.350	0.369	0.361	0.329	0.314	0.257	0.345	
	taud	2.109	2.244	2.340	2.370	2.443	2.494	2.463	2.479	2.551	2.435	2.168	2.598	
	Ebn at Noon	133	206	250	263	274	277	268	260	249	202	124	80	
	Edh at Noon	10	19	26	31	31	30	30	28	21	16	10	6	
All-Sky Solar Radiation	RadAvg	54	237	709	1204	1615	1853	1696	1289	783	327	85	30	
	RadStd	8	45	83	112	158	113	165	100	85	57	18	3	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page